

- 1 What is the number of SI base units required to express electric field strength and magnetic flux density?

	electric field strength	magnetic flux density
A	3	3
B	4	3
C	3	4
D	4	4

Ans: B

$$qE = ma$$

$$[E] = \left[\frac{ma}{It} \right]$$

$$[E] = kg \cdot m \cdot s^{-3} \cdot A^{-1}$$

$$qvB = ma$$

$$[B] = \left[\frac{ma}{Itv} \right]$$

$$[B] = kg \cdot s^{-2} \cdot A^{-1}$$

- 2 The viscosity η of a fluid can be determined by measuring the terminal velocity v of a sphere